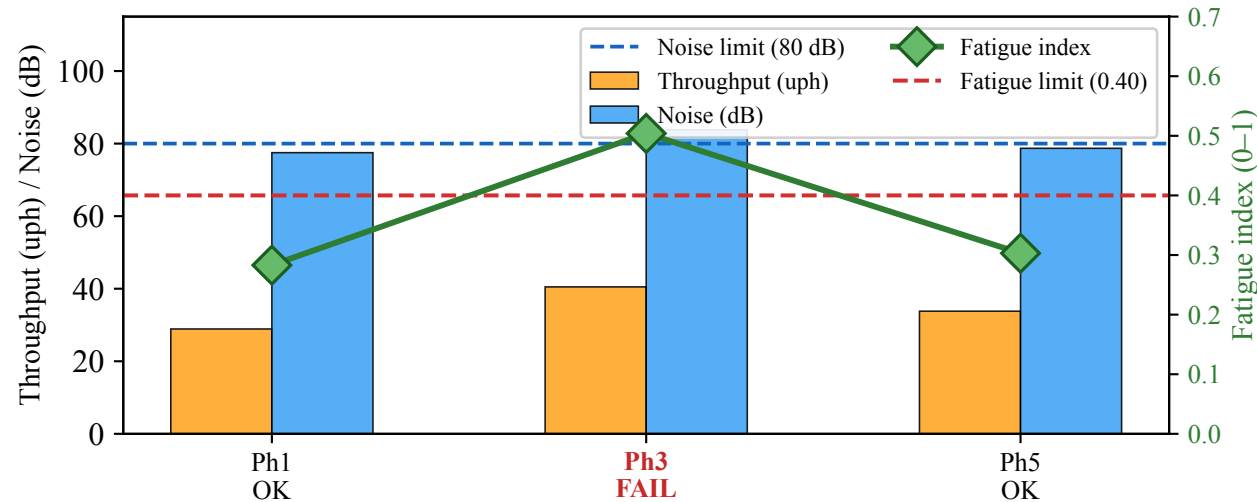
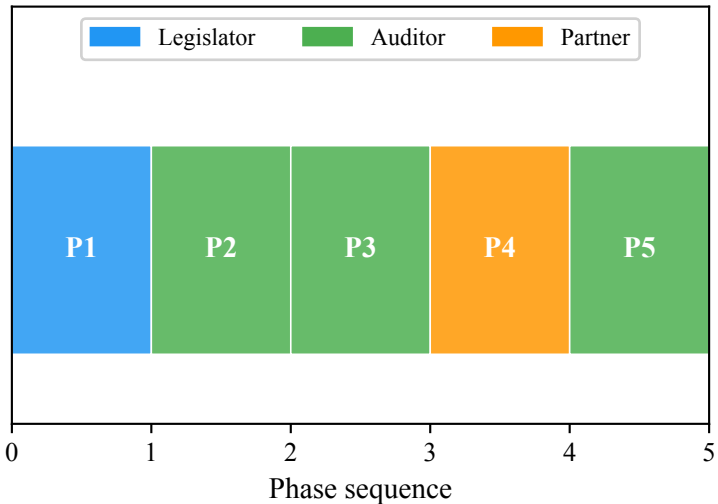


(a) Real-time KPI monitoring with constraint guards



(b) Working-mode lifecycle



(c) OPC-UA node tree exposed to industrial clients

CBPA OPC-UA Server

KPI (Single-cell)

Throughput_uph	33.8
DefectRate	0.008
Noise_dB	78.7
FatigueIndex	0.303
Energy_kWh	374
DeadlineGap_%	35.0

Constraints

FatigueLimit	0.40
NoiseLimit	80.0 dB
CyberRiskLimit	Medium

Contract

Name	C2
Status	ACTIVE

KPI (Factory)

CellBalance_%	1.6
AGV_Util	0.47
Fatigue_H1	0.348
Fatigue_H2	0.0
Fatigue_H3	0.225
FactoryNoise_dB	78.8

(d) Integrated execution stack data flow

